

No. of Functional Features Included in the Solution

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status	Remarks
1	Loan Application Form	Captures 11 applicant attributes through a styled web form	Frontend	Done	Core feature
2	Real-Time Prediction	Runs the saved XGBoost model on submitted data and returns a verdict instantly	Backend / ML	Done	Core feature
3	Data Preprocessing Pipeline	Cleans missing values and encodes categorical features consistently for training and inference	ML Pipeline	Done	Shared logic between notebook and app
4	Class Balancing (SMOTE)	Balances approved vs. rejected classes before training to avoid biased predictions	ML Pipeline	Done	Improves fairness of the model
5	Multi-Model Comparison	Trains and compares Decision Tree, Random Forest, KNN, and XGBoost	ML Pipeline	Done	Selects best performer automatically
6	Animated Result Page	Displays the decision with a stamp-style animation distinguishing Approved vs. Declined	Frontend	Done	Enhances usability and clarity
7	Model Persistence	Saves the trained model and scaler in a portable format for reuse by the Flask app	ML Pipeline	Done	Avoids retraining on every run
8	Cloud Deployment	Hosting the application on a public server (e.g., IBM Cloud)	Deployment	Pending	Currently runs on local Flask server

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home page (passbook-style landing), loan application input form with live progress meter, animated Approved / Declined result page with stamp animation
2	Backend / Logic	4	Flask routing (/ , /predict, /submit), form input preprocessing and encoding, StandardScaler transformation, XGBoost model inference and result rendering
3	Database / Storage	2	Trained model saved as rdf_model.json (XGBoost native format), fitted scaler saved as scale1.pkl; training dataset stored as loan_prediction.csv
4	API / Integration	1	Flask POST endpoint (/submit) that accepts form data, preprocesses it, and returns a prediction response — no external third-party APIs used
5	Security / Authentication	0	Not implemented in this iteration. HTML5 required-field validation prevents empty submissions. Future scope: role-based login for credit officers vs. applicants